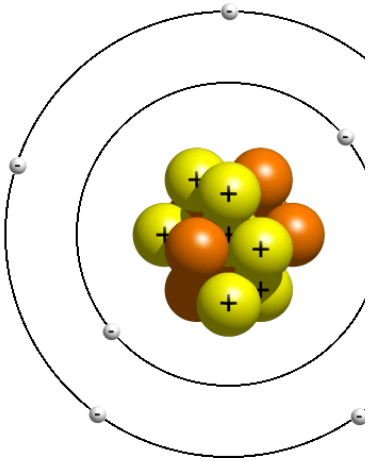
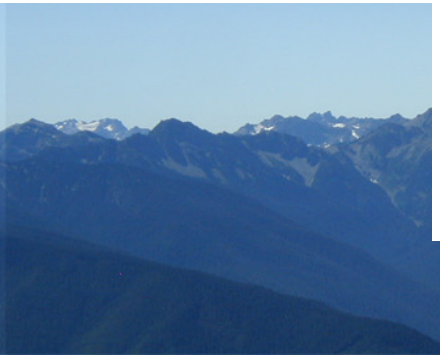
When Physics and Computer Science Collide

Dave Bacon
Department of Computer Science & Engineering
University of Washington, Seattle, WA

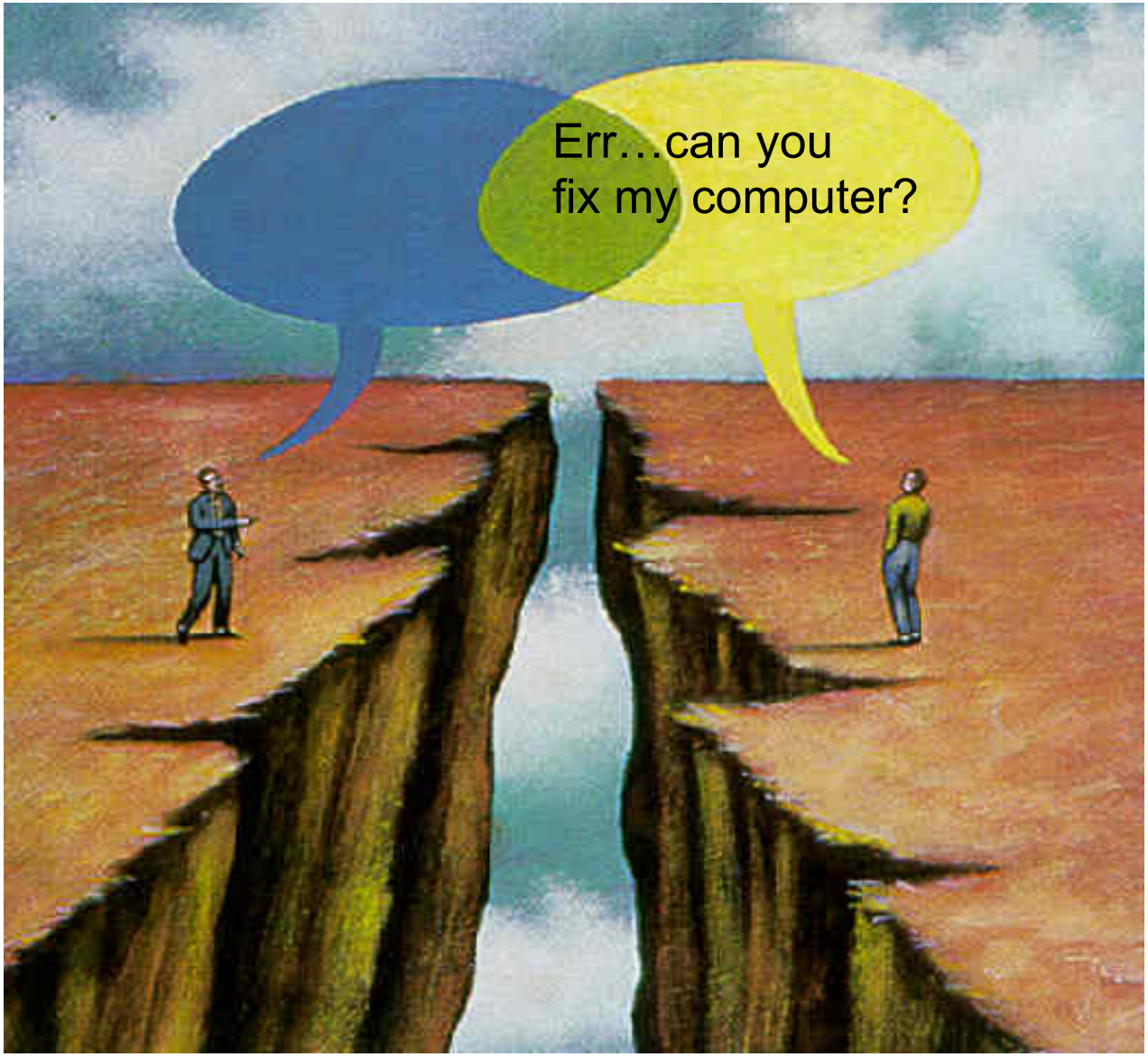


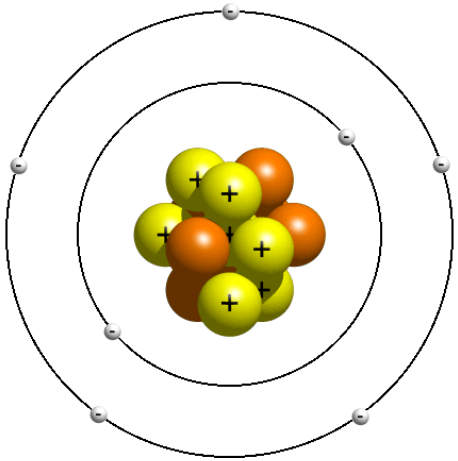






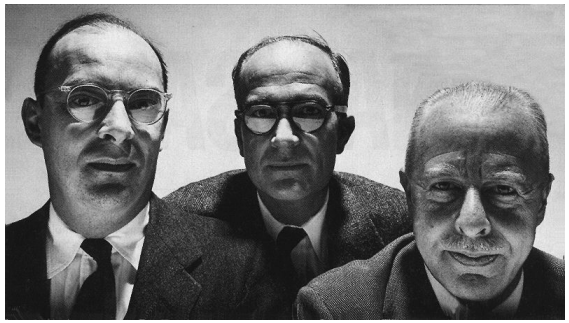



Err…can you
fix my computer?

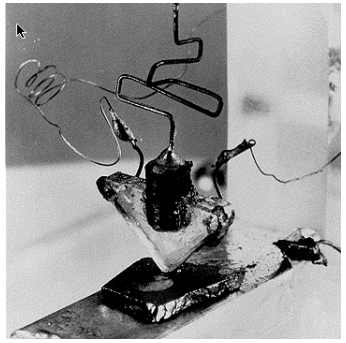




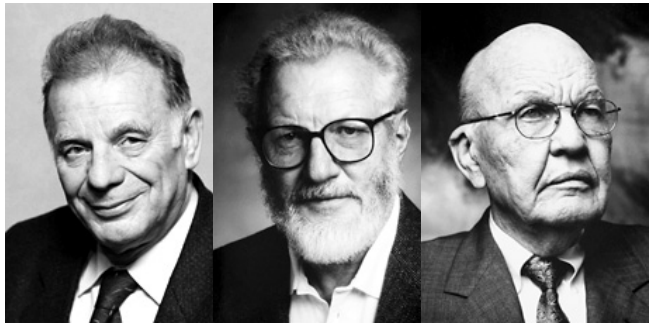




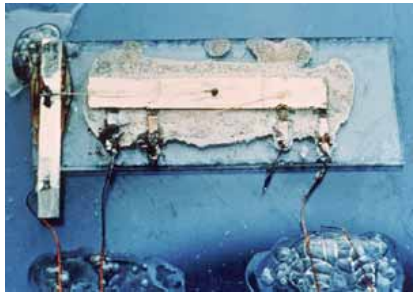
1956
Shockley, Bardeen, and Brattain



Transistor

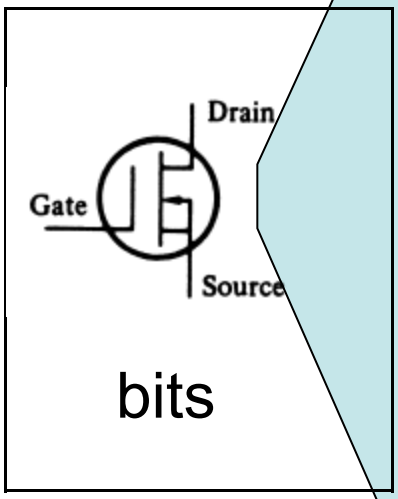


2000
Alferov, Kroemer, Kilby

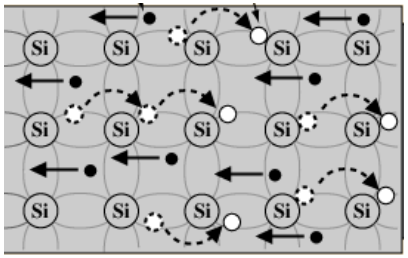


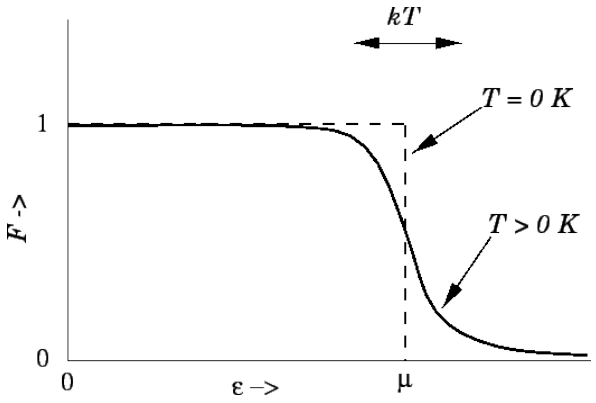
Integrated Circuit

Behind the Bit

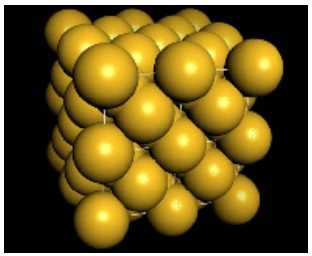


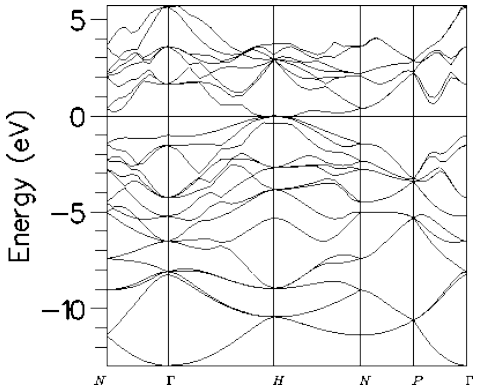
bits





noisy bits





quantum bit

Emergence of orders which can compute

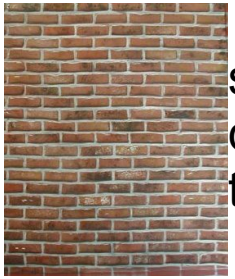
Quantum Computer?



Richard Feynman

"How can we simulate quantum mechanics? ...Can we do it with a new kind of computer… a quantum computer? It's not a Turing machine but a machine of a different kind" - Richard Feynman (1981)





simulating quantum theory

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

FACTORING Classical: $O(\exp(n^{1/3}))$ Quantum: $O(n^2 \log n)$

Peter Shor 94

Quantum Computers!



Public Key Cryptosystems

RSA **BROKEN**
Diffie-Hellman
Elliptic Curves



188198812920607963838697239461650439807163563379417382700763356422988859715234654853190606065047430453173880113033967161969232120573403187955065699622130516875930765025705 9

"The problem of distinguishing prime numbers from composites, and of resolving composite numbers into their prime factors, is one of the most important and useful in all of arithmetic... The dignity of science seems to demand that every aid to the solution of such an elegant and celebrated problem be zealously cultivated"
— Carl Gauss



Gauss

Quantum Computers!

Quantum algorithms that offer speedups over classical algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- Simulation of quantum systems



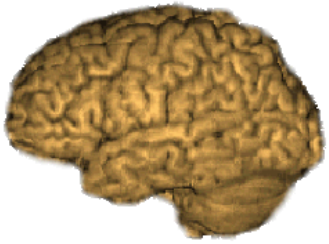
Applications to chemistry, biology, material science….

But Can We Build Quantum Computers?

Two Views of Information

Information is "fungible" adj. being of such a nature that one part or quantity may be replaced by another equal part or quantity in the satisfaction of an obligation.







Al Gore

George Bush

Ralph Nader

Pat Buchanan

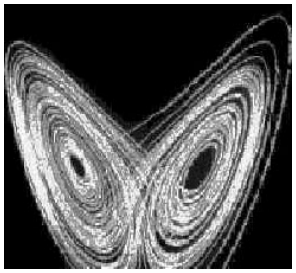


No Information without Physical Representation!



Rolf Landauer

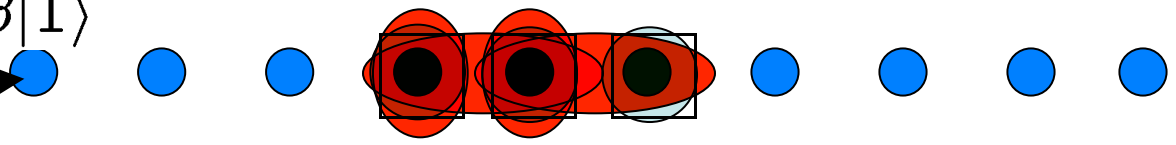
Not all physical systems are robust information processing machines



The Ideal Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

qubits



$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$

Quantum algorithm

1. Initialize
2. Series of unitary gates
3. Measurement

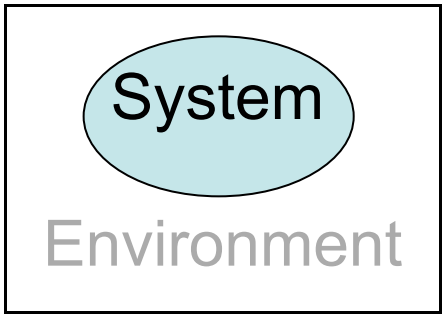
0

1

1

Obstructions

DECOHERENCE



Total Hamiltonian

$$H = H_S \otimes I + I \otimes H_E + H_I$$

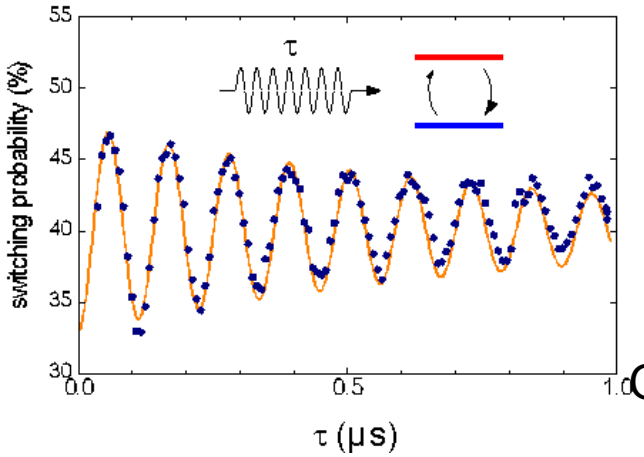
system environment interaction

open quantum systems can become *entangled* with their environment

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

coherent superposition incoherent sum

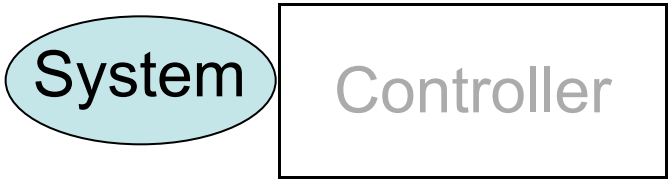
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$



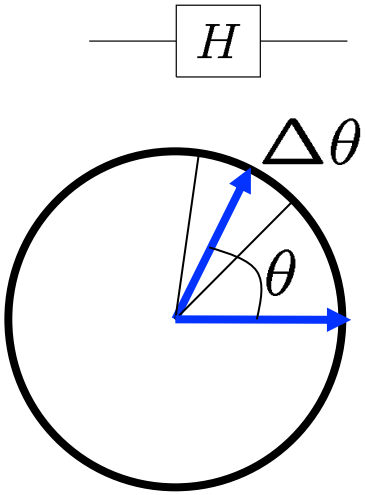
Quantronium (CEA, France)

More Obstructions

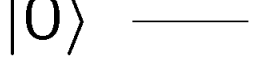
FAULTY CONTROL



Imprecise or faulty unitary gates

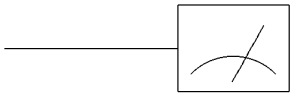


Imperfect state preparation



$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

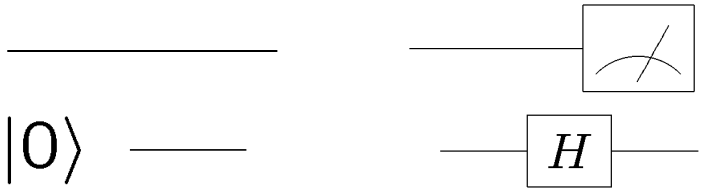
Imperfect measurements



$$\alpha|0\rangle + \beta|1\rangle \begin{cases} \rightarrow Pr(0) = |\alpha|^2 - \epsilon \\ \rightarrow Pr(1) = |\beta|^2 + \epsilon \end{cases}$$

This is the End, My Only Friend, the End

Everything, (quantum) noisy, all the time.



Is a Quantum Computer Even a Computer?

A skeptics checklist:



Laughlin

1. quantum data is analog
2. quantum data likes to become classical data
3. quantum evolution is analog
4. quantum theory is wrong

Quantum Data is Analog

n qubit quantum state: $|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle$



Objection:

Superposition of an exponential number of states.

Each amplitude is a complex number…i.e. analog data

How do you control 2n complex numbers?

In the real world, nothing can be done with infinite precision

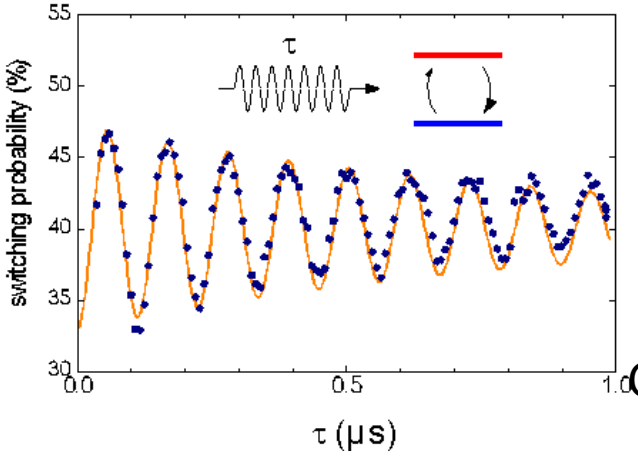
Quantum data likes to become classical data

coherent superposition

incoherent sum

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2|0\rangle_S\langle 0|_S + |\beta|^2|1\rangle_S\langle 1|_S$$

Decoherence is a complex process which is fundamentally responsible for there being classical information



Quantronium (CEA, France)

Decoherence is irreverislbe

Quantum Evolution is Analog

Quantum evolution: $U(t) = \exp(-iHt)$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

A unitary “gate” also has a bunch of complex numbers

Each amplitude is a complex number…i.e. analog data

In the real world, nothing can be done with infinite precision

Quantum Theory is Wrong

Quantum theory has only ever been observed for small (micro to meso) systems.

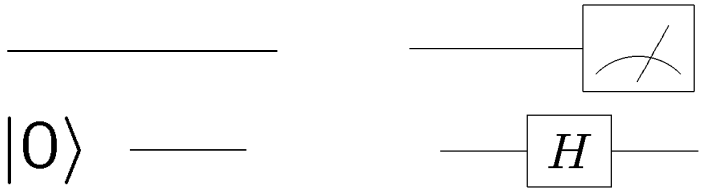
It will break down for large systems



No Job For You Corollary: Do not think to hard about quantum theory or you will get lost in a philosophical hole from which no graduate student has ever emerged (except in philosophy departments)

This is the End, My Only Friend, the End

Everything, (quantum) noisy, all the time.



Is a Quantum Computer Even a Computer?

A skeptics checklist:

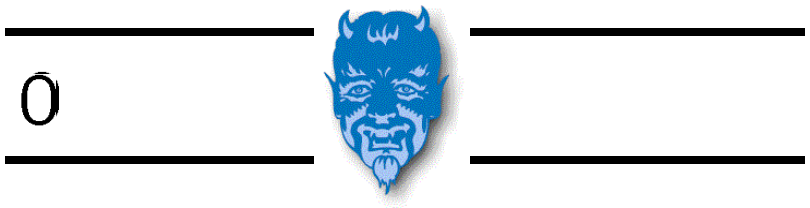


Laughlin

1. quantum data is analog
2. quantum data likes to become classical data
3. quantum evolution is analog
4. quantum theory is wrong

Why Is Classical Computation Possible?

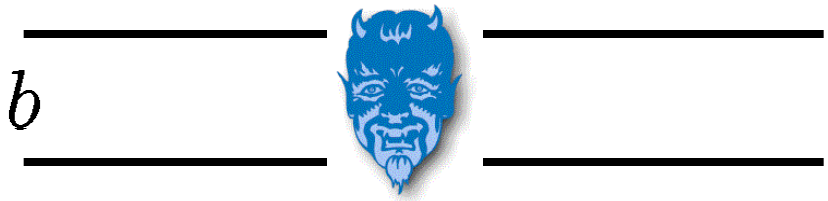
Classical Solutions



Probability of bit flip = *p*

Solution: redundancy

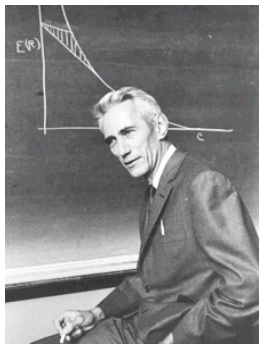
$0 \rightarrow 000$
$1 \rightarrow 111$



$$\left.\begin{matrix} 000 \\ 001 \\ 010 \\ 100 \end{matrix}\right\} \rightarrow 0$$

$$\left.\begin{matrix} 111 \\ 110 \\ 101 \\ 011 \end{matrix}\right\} \rightarrow 1$$

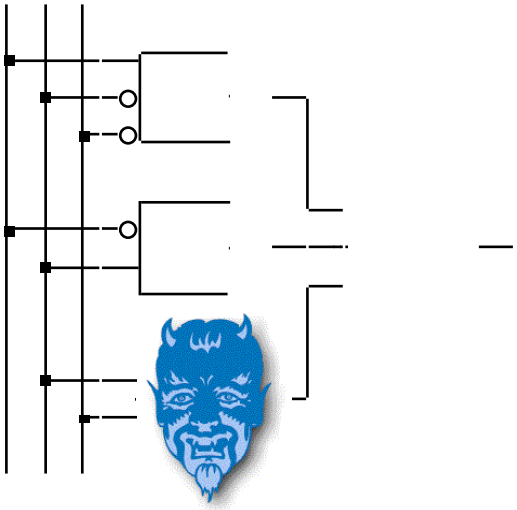
Probability of failure =*3(1-p)p2+p3*
=*3p2-2p3*
<*p* if *p*<*1/2*



Use redundancy to protect information against noise

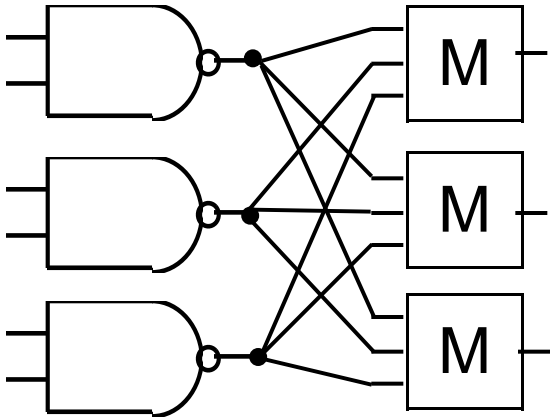
Claude Shannon 1948
“A mathematical theory of communication”

More Classical Solutions



Classical gates fail with probability p

Solution: error correcting codes + fault tolerant construction



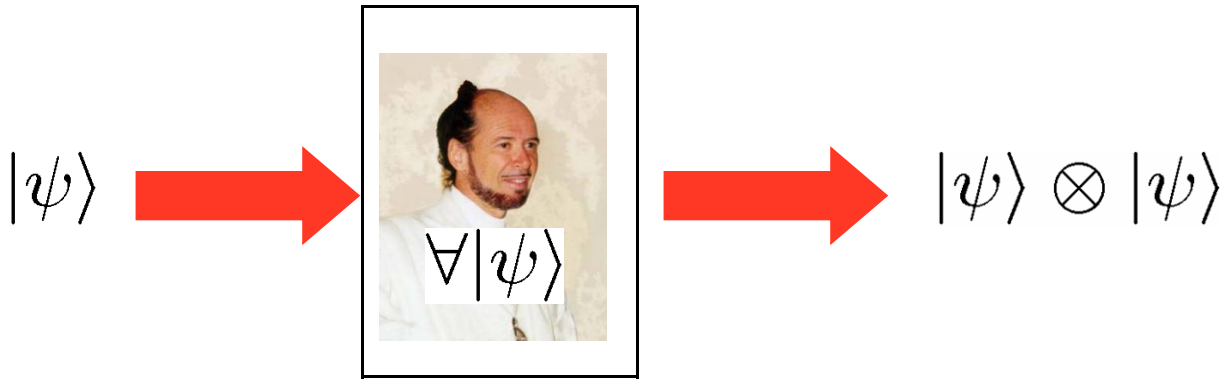


John von Neumann, 1956

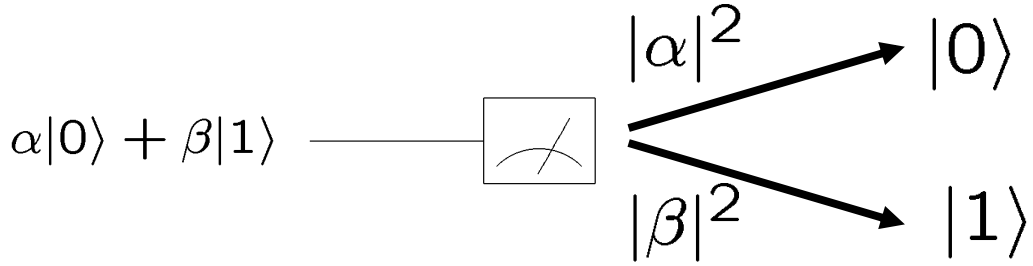
“Probabilistic logics and the synthesis of reliable organisms from unreliable components”

Quantum Problems?

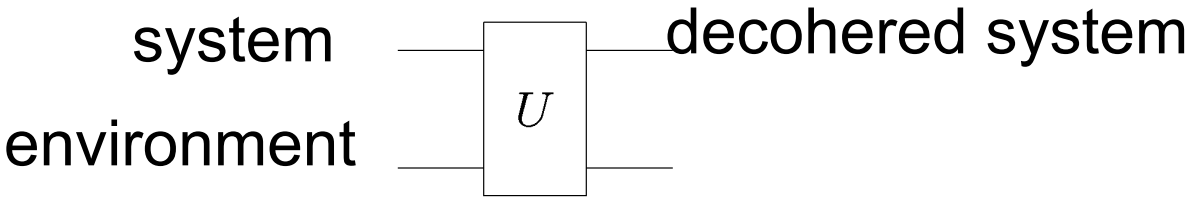
No cloning of quantum information



How to perform error correction without measuring?

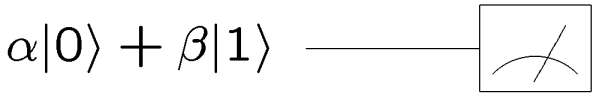


Quantum errors seem “analog”

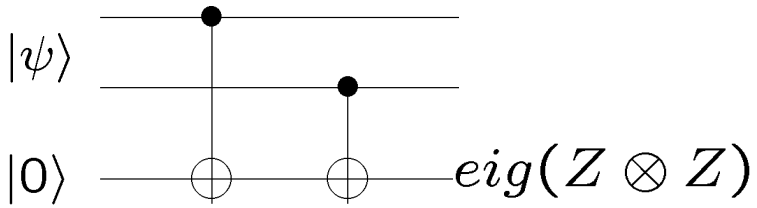


Quantum Observations

How to perform error correction without measuring?



Observation: not all measurements destroy quantum information



$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

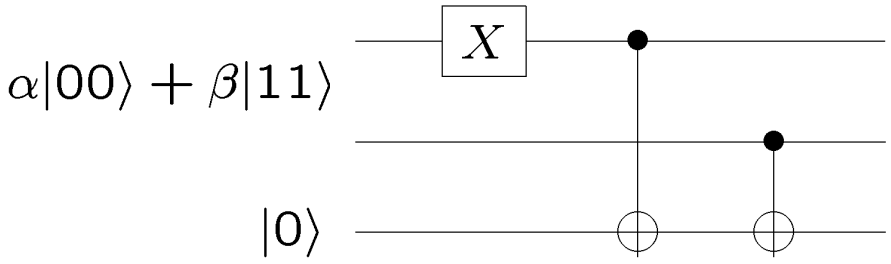
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

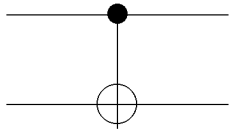
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Information encoded into a subspace undisturbed

$$\alpha|00\rangle + \beta|11\rangle \rightarrow \alpha|00\rangle + \beta|11\rangle \qquad eig(Z \otimes Z) = +1$$

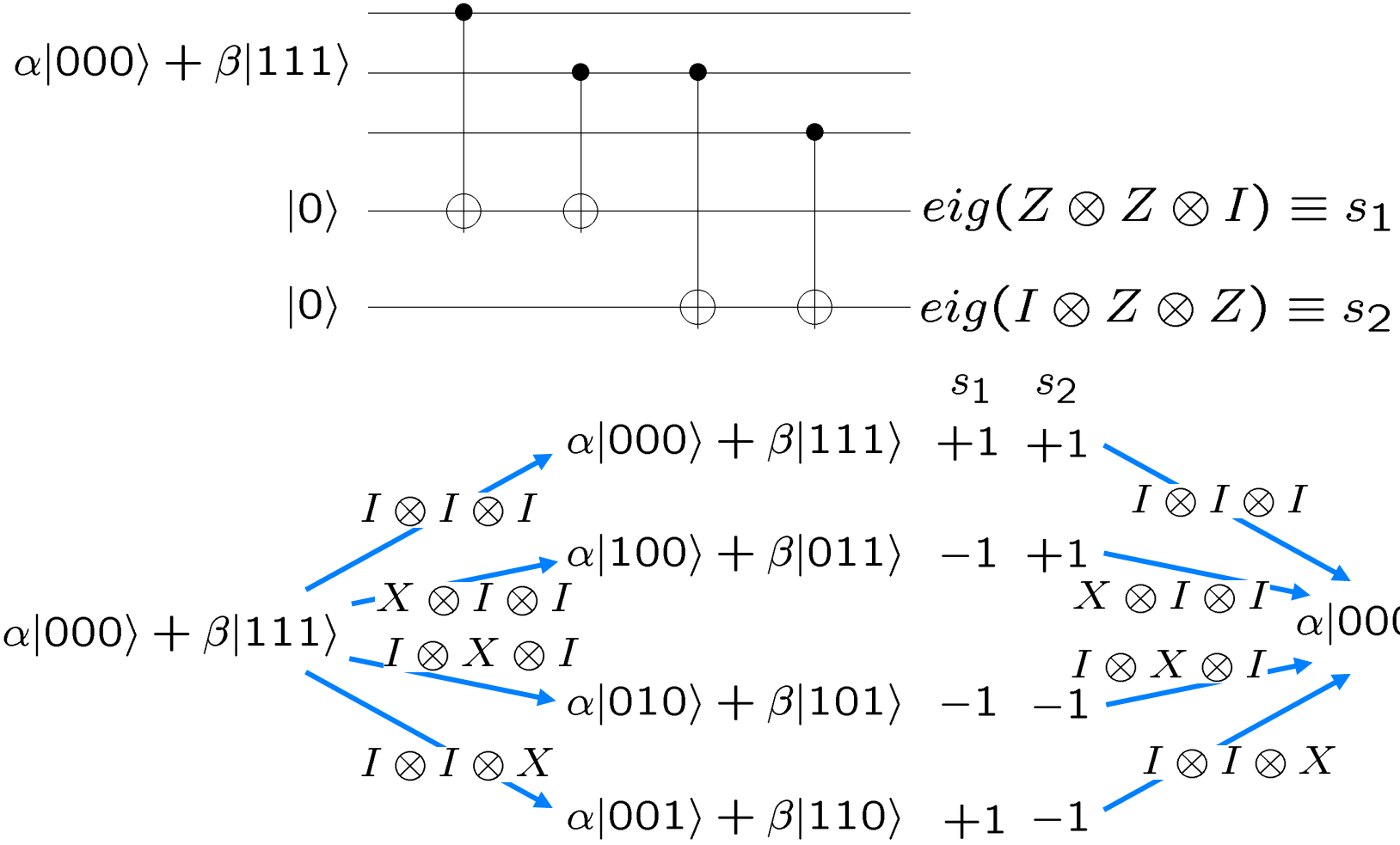
Can detect “leaving” of subspace





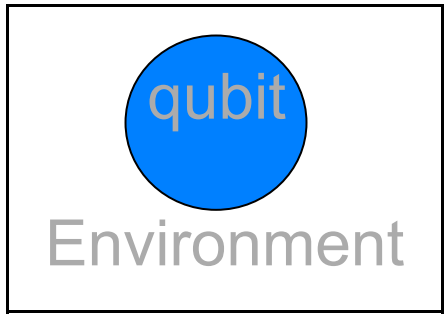
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Quantum Observations

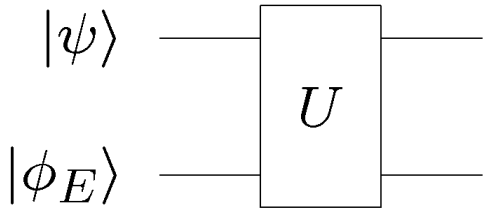


By encoding into subspaces we can correct single bit flips

Quantum Problems?



$$H = H_S \otimes I + I \otimes H_E + H_I$$



$$|\psi\rangle\otimes|E_I\rangle+(X|\psi\rangle)\otimes|E_X\rangle+(Y|\psi\rangle)\otimes|E_Y\rangle+(Z|\psi\rangle)\otimes|E_Z\rangle$$

digitized effects of decoherence

Think: amplitude for no error, *X* error, *Y* error, *Z* error

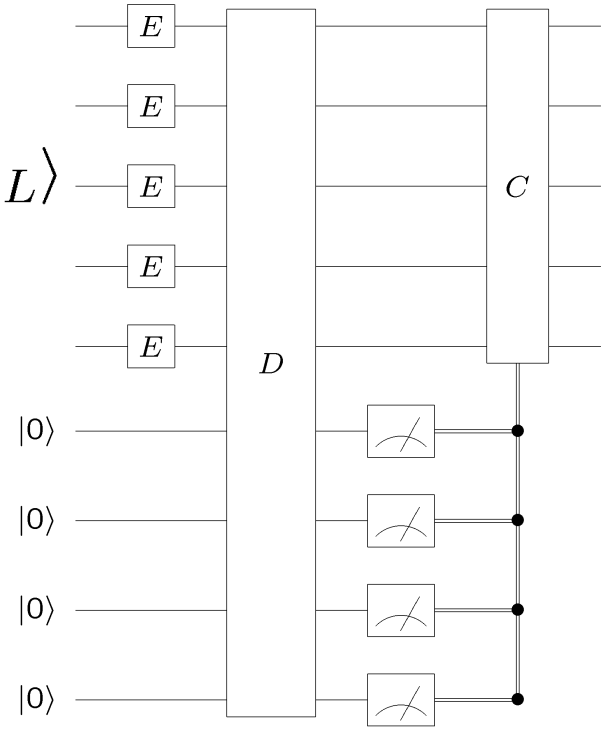
Quantum Error Correcting Codes

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I \qquad [s_i, s_j] = 0$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

Encode qubit into 2 dimensional subspace with all +1 eigenvalues

Single qubit errors (i.e. $I \otimes X \otimes I \otimes I \otimes I, I \otimes I \otimes Y \otimes I \otimes I$) all lead to a unique subspace labeled by *si* eigenvalues

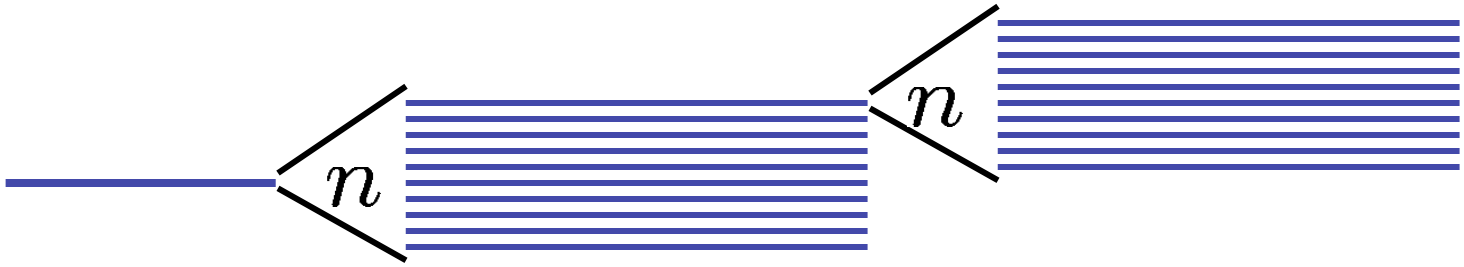
$\alpha|0_L\rangle + \beta|1_L\rangle$



$\approx \alpha|0_L\rangle + \beta|1_L\rangle$

"bare" error rate = *p*
error rate on encoded = *O(p2)*

Perfection Through Concatenation



failure probability

p *cp2* *c(cp2)2*

At kth level of concatenation, probability of an error which destroys the encoded quantum information is

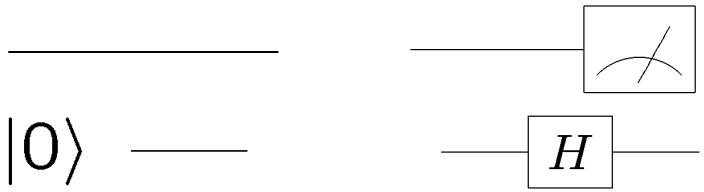
$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}}\left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log\frac{1}{p_{fail}}\right)$$

Fault-Tolerance

Everything noisy:



Compute on qubits encoded in quantum error correcting code?

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

encoded Paulis

$$\bar{X} = X \otimes X \otimes X \otimes X \otimes X$$
$$\bar{Y} = Y \otimes Y \otimes Y \otimes Y \otimes Y$$
$$\bar{Z} = Z \otimes Z \otimes Z \otimes Z \otimes Z$$

COMPARE

$$\exp\left[-itZ \otimes Z \otimes Z \otimes Z \otimes Z\right] = \cos(t) I - i \sin(t) \bar{Z}$$

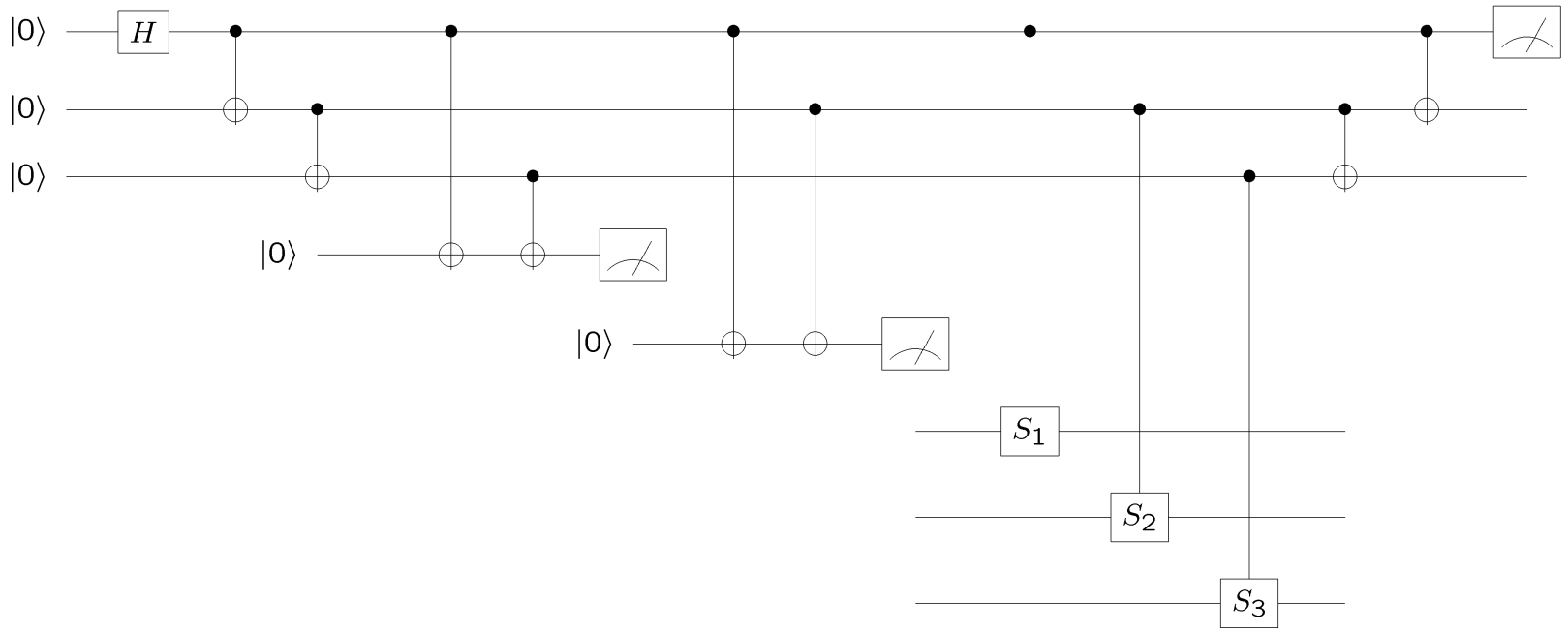
$t \approx \frac{\pi}{2}$ Exact: $-i\bar{Z}$ Error: $\bar{Z}$

$$\exp\left[-itZ\right] \otimes \exp\left[-itZ\right] \otimes \exp\left[-itZ\right] \otimes \exp\left[-itZ\right]$$

$t \approx \frac{\pi}{2}$ Exact: $-i\bar{Z}$ Error: single qubit Z

$$Z \otimes I \otimes I \otimes I \otimes I$$

Fault-Tolerant Quantum Computation



Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, measurements) can fail.

Quantum error correction + Fault-tolerant procedures

Accuracy?



Error in probabilities is less than

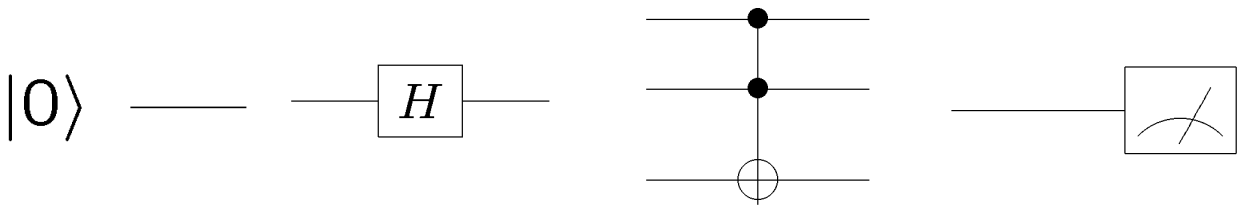
$$E(U,V) = \max_{|||\psi\rangle||^2=1} ||(U-V)|\psi\rangle||$$



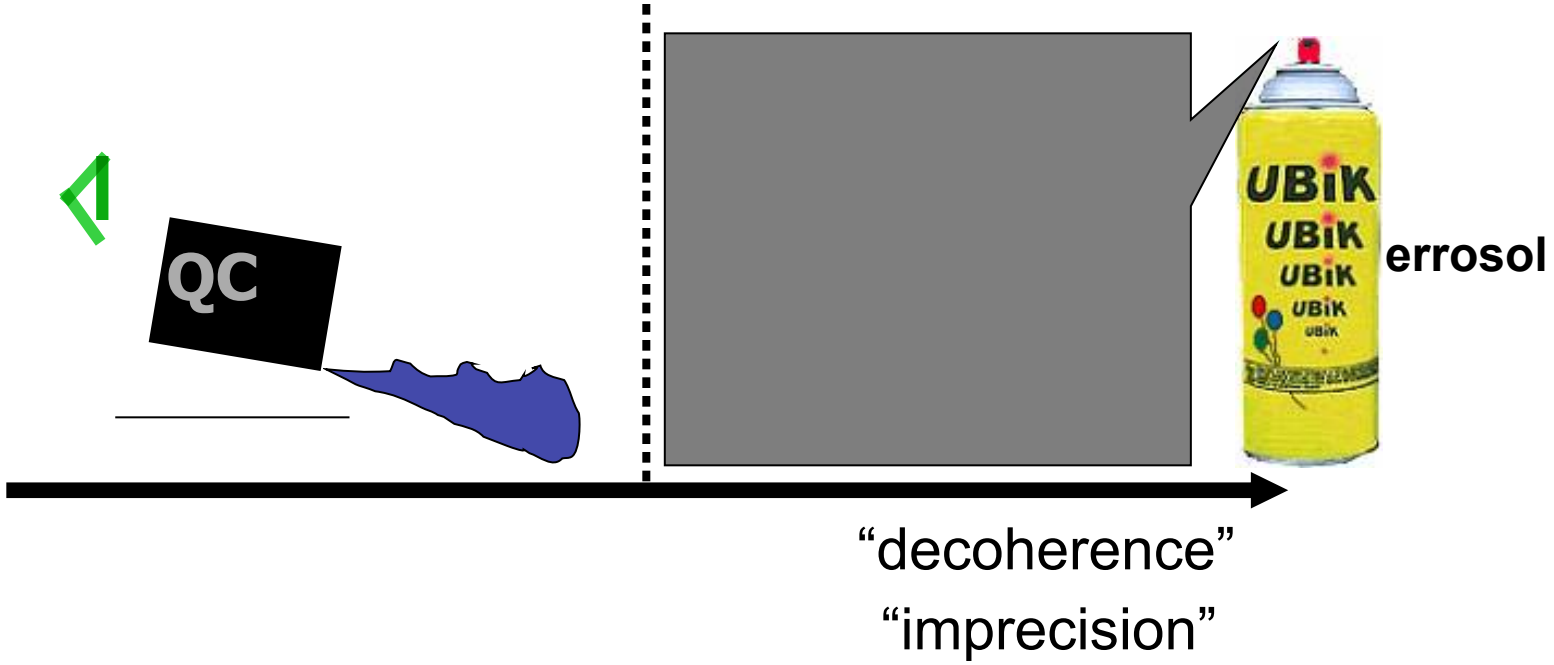
Errors <u>add</u> $E(U_m \cdots U_3 U_2 U_1, V_m \cdots V_3 V_2 V_1) \leq \sum_{i=1}^{m} E(U_i, V_i)$

Each gate should be implemented with polynomial accuracy in circuit size

Fault-tolerant universal gate sets:



Quantum Threshold Theorem



Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev, Gottesman, Preskill, Shor, + others]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Lessons

1. quantum data is analog

Degrees of freedom	Description of system
analog	deterministic
digital	probabilistic
	quantum

2. quantum data likes to become classical data

quantum information can be encoded
encodings can reduce effective error rate
quantum noise can be “digitized”

3. quantum evolution is analog

encoded quantum information can be computed on
fault-tolerant constructions can be error corrected

4. quantum theory is wrong

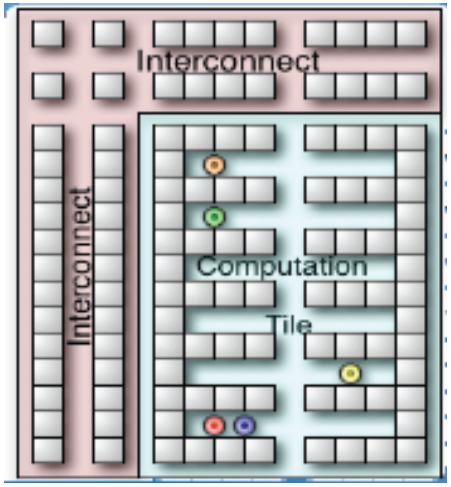
maybe, but then…what a discovery!!!

Lessons

No Job For You Corollary: Do not think to hard about quantum theory or you will get lost in a philosophical hole from which no graduate student has ever emerged (except in philosophy departments)

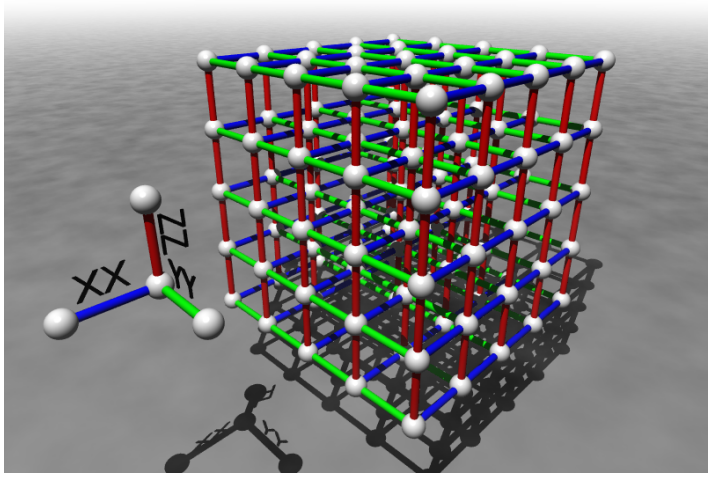
Quantum information science is not about questioning quantum theory but about **accepting it!**

Our Probable Paths



quantum error correcting architectures

Engineering



Physics motivated methods for error correction

Physics

Threshold(s) Theorem(s)

Threshold theorems come in many different forms….

provable bounds, Monte Carlo estimates, …

different shapes…

assumptions of error models
assumptions of physical devices
assumptions of architectures

yielding threshold values for relative error rates from
10-9 to 10-2

…but device complexities are universally very high.
(hundreds to thousands to millions of “bare” qubits per encoded qubit with extremely complex manipulations)

History

1994-5 Shor, Steane invent quantum error correction

Shor invents FT protocols

1996

Aharonov, Ben-Or
FT Proof

Knill, Laflamme, Zurek
FT Estimate

Zalka
Threshold estimate
(10-4)

Kiteav
Topological QC
FT Proof

1997-2003 Everyone admires how cool the threshold theorem is

2004-NOW

Renaissance

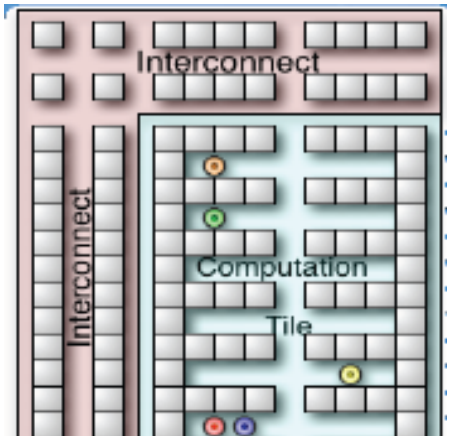
Aliferis, Gottesman, Preskill
FT Proof (2.7 x 10-5)

Aliferis, Cross
FT Proof (1.9 x 10-4)

Knill / Reichardt (1 x10-2)
Threshold estimates
(High overhead ancilla factories)

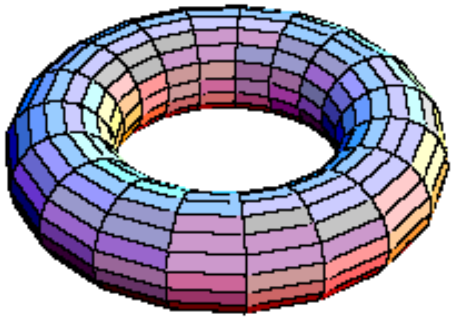
Developments

Spatial architecture



Gottesman / Aharonov, Ben-Or (1999)
Still a threshold with “local gates”

Denis, Gottesman, Kitaev, Preskill (2001)
Topological quantum memory
Memory threshold of 10-4
(Fast reliable classical control)



Svore, Terhal, DiVincenzo (2004)
~factor of 10 loss in 2D

Szkopek et al., (2004)
~factor of 100 loss in 1D

Raussendorf, Harrington (2006)
Local in 2D, threshold estimate 5x10-3

Developments, II

Is the model of noise correct?

Alicki, Horodecki, Horodecki, Horodecki (2001)

Alicki, Lidar, Zanardi (2005)

Independent bath model. Non-Markovian noise?



Terhal, Burkhard (2004)

Separate environments, non-Markovian noise

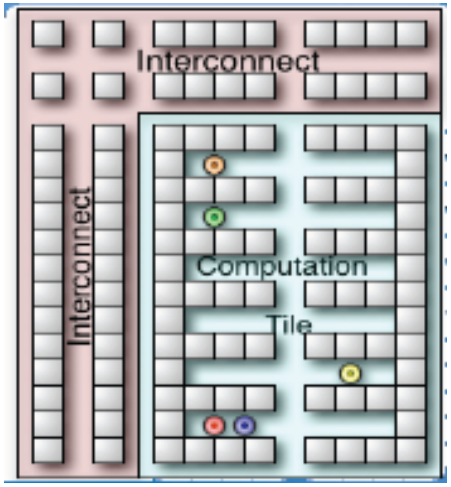
Aliferis, Gottesman, Preskill (2005)

Shared environments, non-Markovian noise

Aharonov, Kitaev, Preskill (2005)

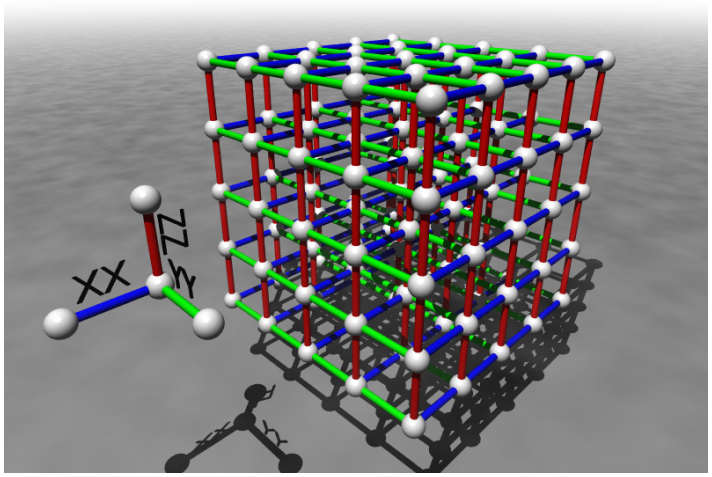
Algebraically decaying correlated noise

Our Probable Paths



quantum error correcting architectures

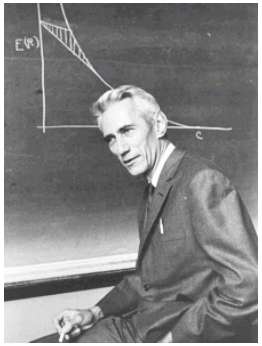
Engineering



Physics motivated methods for error correction

Physics

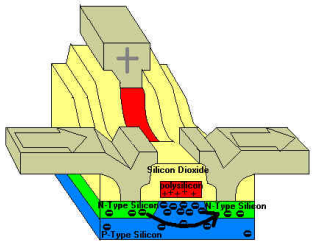
Back To Reality





theory

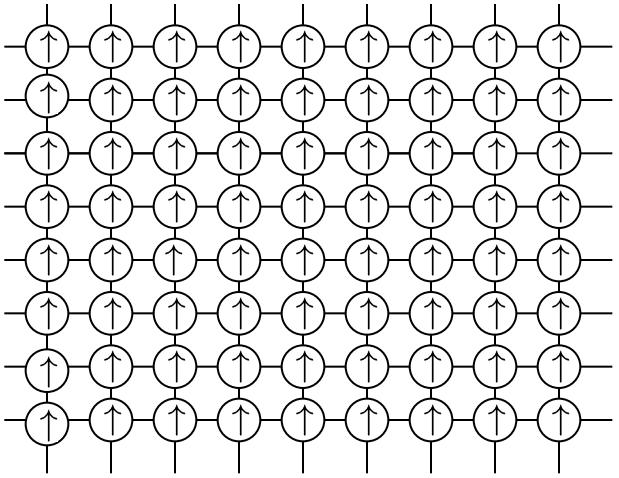




practice

Practical devices of information processing enact the ideas of fault-tolerant classical computation in the *physics* of the devices.

Toy Model

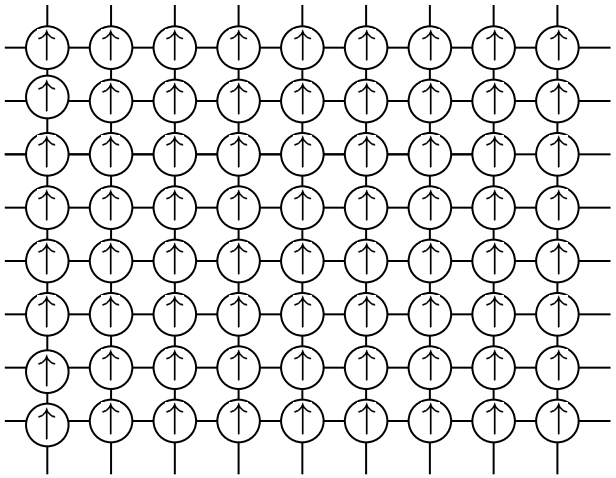


Ising model in two+ dimensions

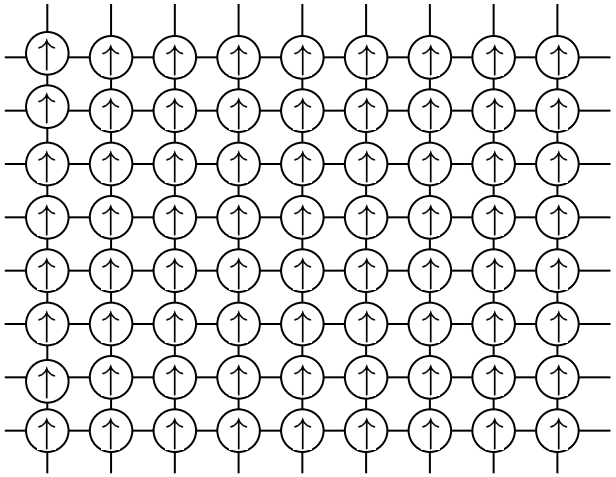
$$H = -J \sum_{\langle i,j \rangle} s_i s_j$$

Redundancy code:

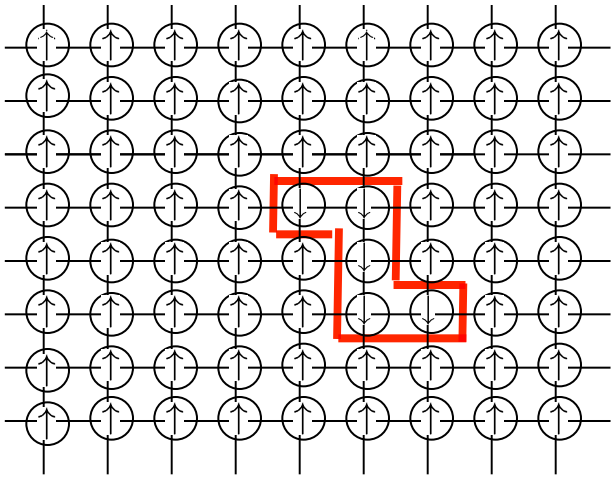
$0 \rightarrow$



$1 \rightarrow$

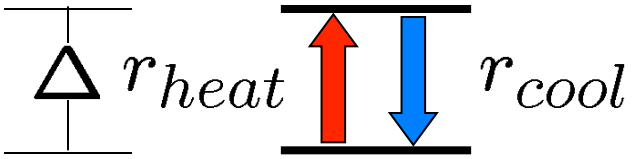


Dynamical Ising Model



Energy of excitation=
2J (Sum of perimeters of flipped domains)

Dynamics obeying principle of detailed balance:



$$\frac{r_{heat}}{r_{cool}} = e^{-\frac{\Delta}{k_B T}}$$

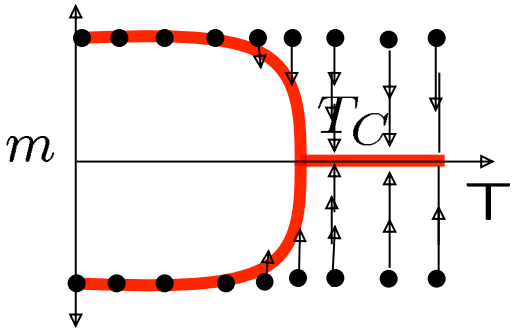
Competition between growth and decimation of flipped domains



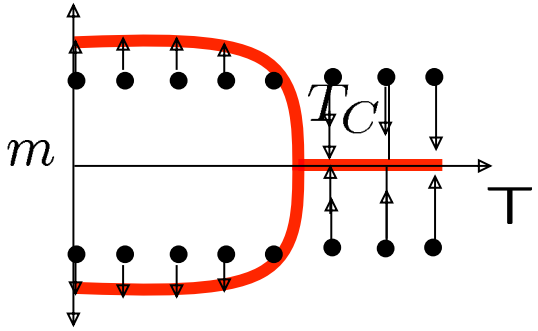
Robust



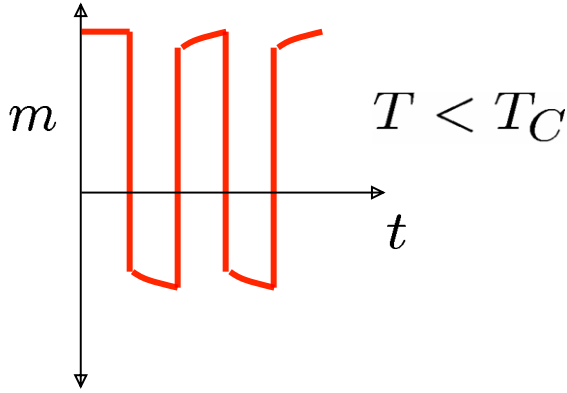
Relaxation to thermal



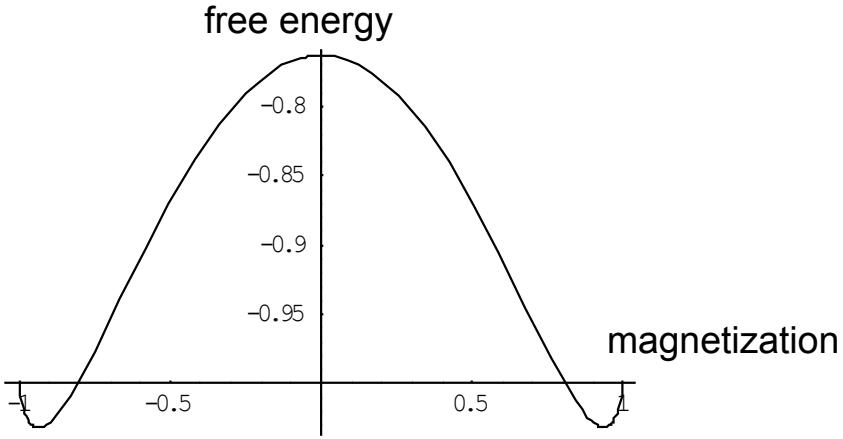
Imperfect preparation



Manipulation

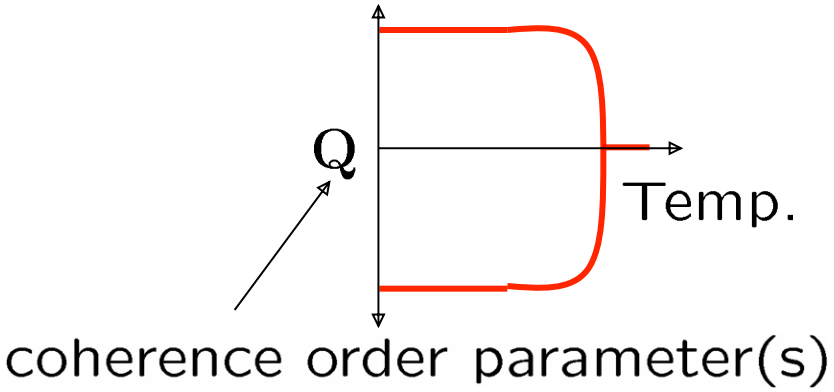


The two encoded states have lifetimes which grow with the size of the system:



Quantum Hard Drive?

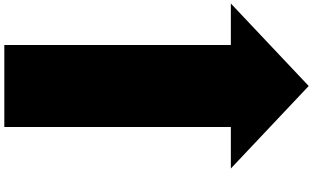
Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



Today in QC: we have theory (like von Neumann, Shannon) but not devices which implement these idea (without severe device complexity overhead.)

Quantum Threshold

Threshold Theorem



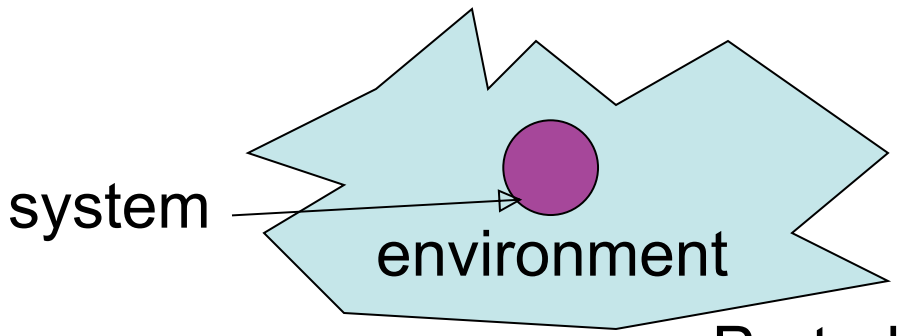
New type of order in quantum many-body systems

4 spatial dimensional toric code is a quantum hard drive
[Dennis, Gottesman, Kitaev, Preskill 2001]

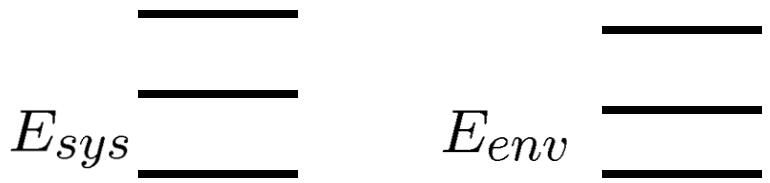
Problems:

- 4 spatial dimensions!
- Hamiltonian involves 6 and great qubit interactions

Energetics of Decoherence

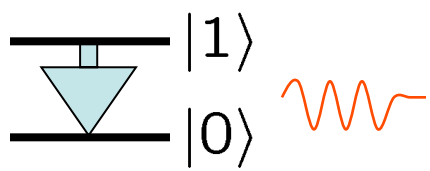


$$H = H_{sys} + H_{int} + H_{env}$$



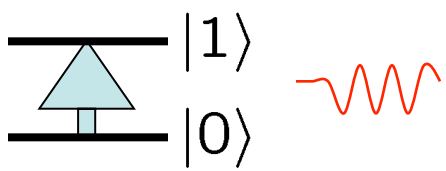
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy (rotating wave approx)



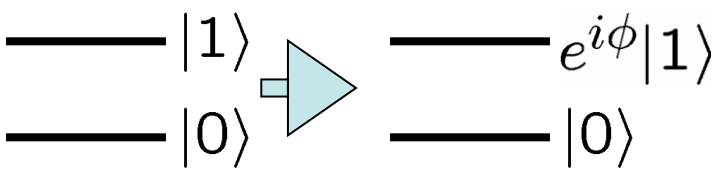
cooling

$\Delta E_{sys} < 0$



heating

$\Delta E_{sys} > 0$

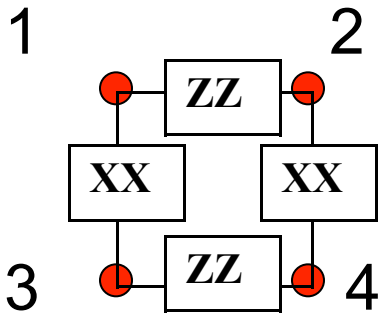


dephasing

$\Delta E_{sys} = 0$

by **cooling environment** we can eliminate heating

1st Attempt



quantum

Hamiltonian

$$H = -(Z_1Z_2 + Z_3Z_4 + X_1X_3 + X_2X_4)$$

Quantum error detecting code

$$S_1 = Z_1Z_2Z_3Z_4$$
$$S_2 = X_1X_2X_3X_4$$
$$S_i|\psi_C\rangle = |\psi_C\rangle$$
$$\bar{Z}_1 = Z_1Z_2 \qquad \bar{X}_1 = Z_1Z_3$$
$$\bar{Z}_2 = Z_1Z_3 \qquad \bar{X}_2 = X_1X_2$$

Code is a symmetry of the Hamiltonian:

$$[H, S_1] = [H, S_2] = 0$$

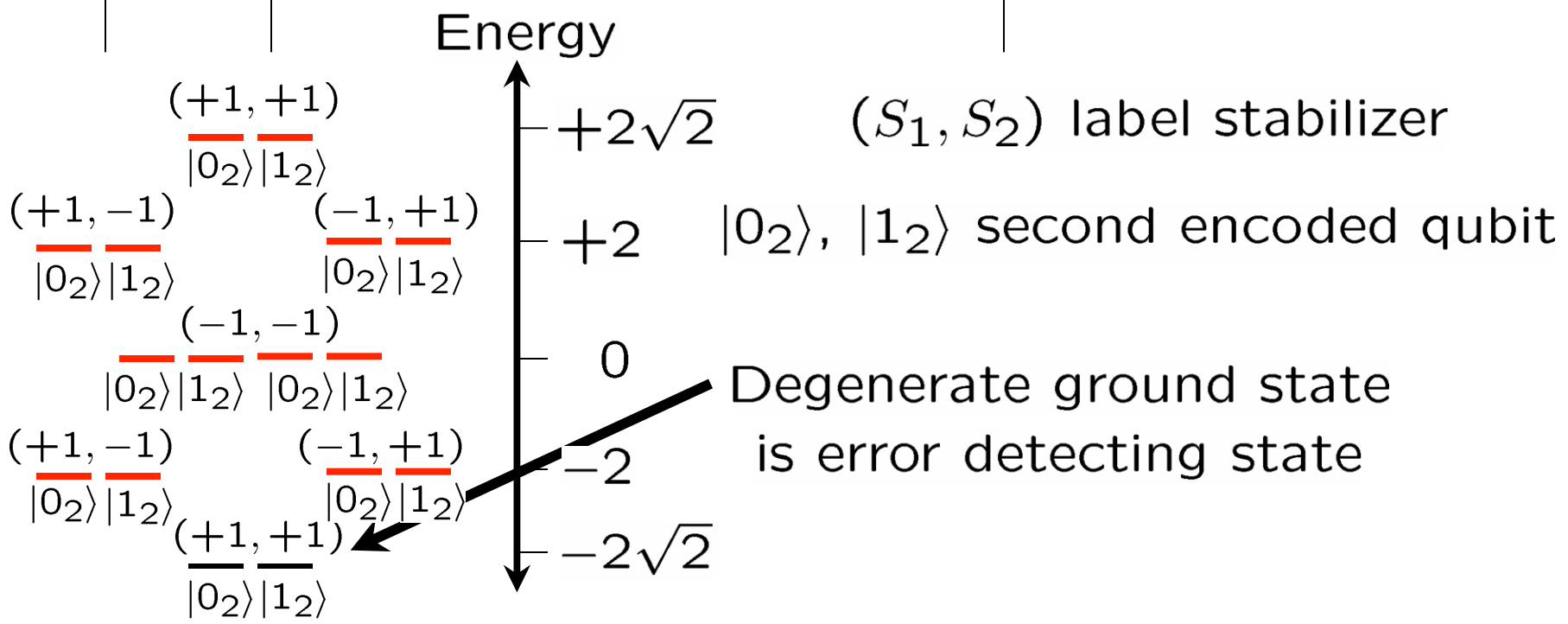
Code detects any single qubit error:

$$\{P, S_i\} = 0 \qquad P \text{ e.g. } Q \otimes I \otimes I \otimes I$$

H does not depend on second qubit (use to store encoded qubit)

$$H = -\bar{Z}_1(I + S_1) - \bar{X}_1(I + S_2)$$

S_1	S_2	$H = \bar{X}_1(1+S_1) + \bar{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\bar{X}_1 + \bar{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\bar{X}_1$	± 2
-1	$+1$	$2\bar{Z}_1$	± 2
-1	-1	0	0

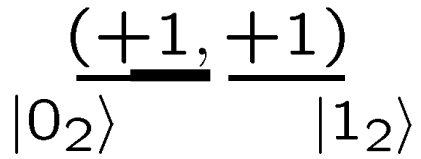


ground state is +1 eigenvalue of four qubit interactions

Supercoherence

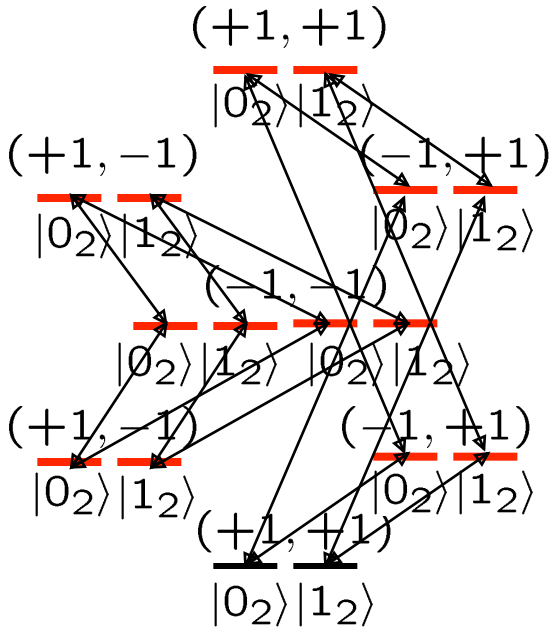
ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.



FORBIDDEN

Effect of **ZIII**

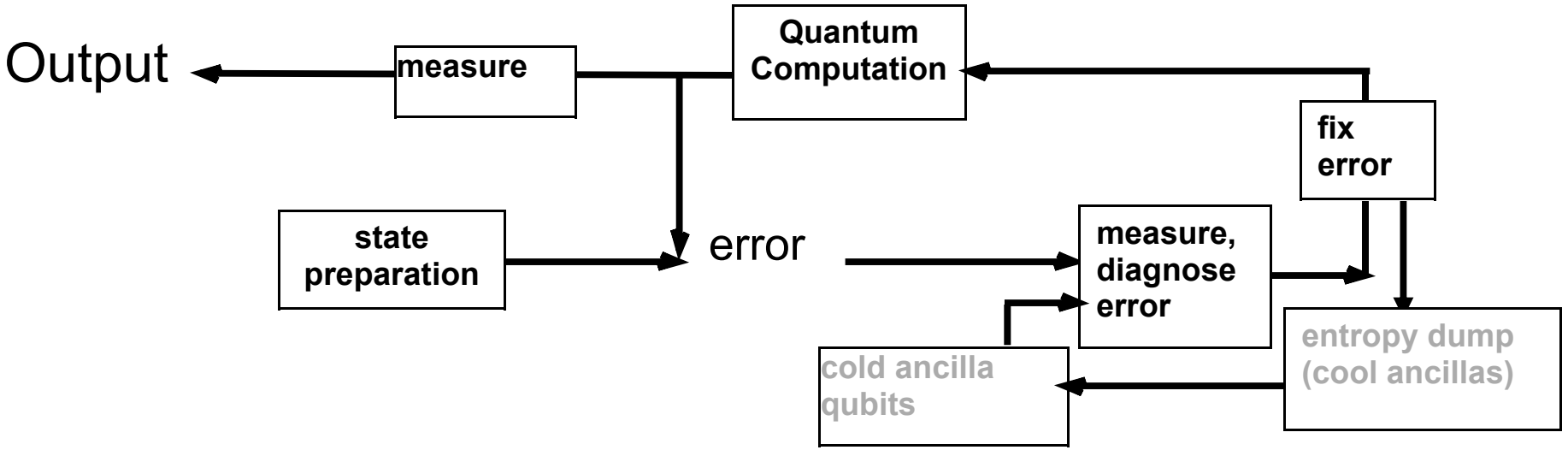


To first order all single qubit decoherence becomes heating

Energy gap and low temperature suppress 1st order decoherence

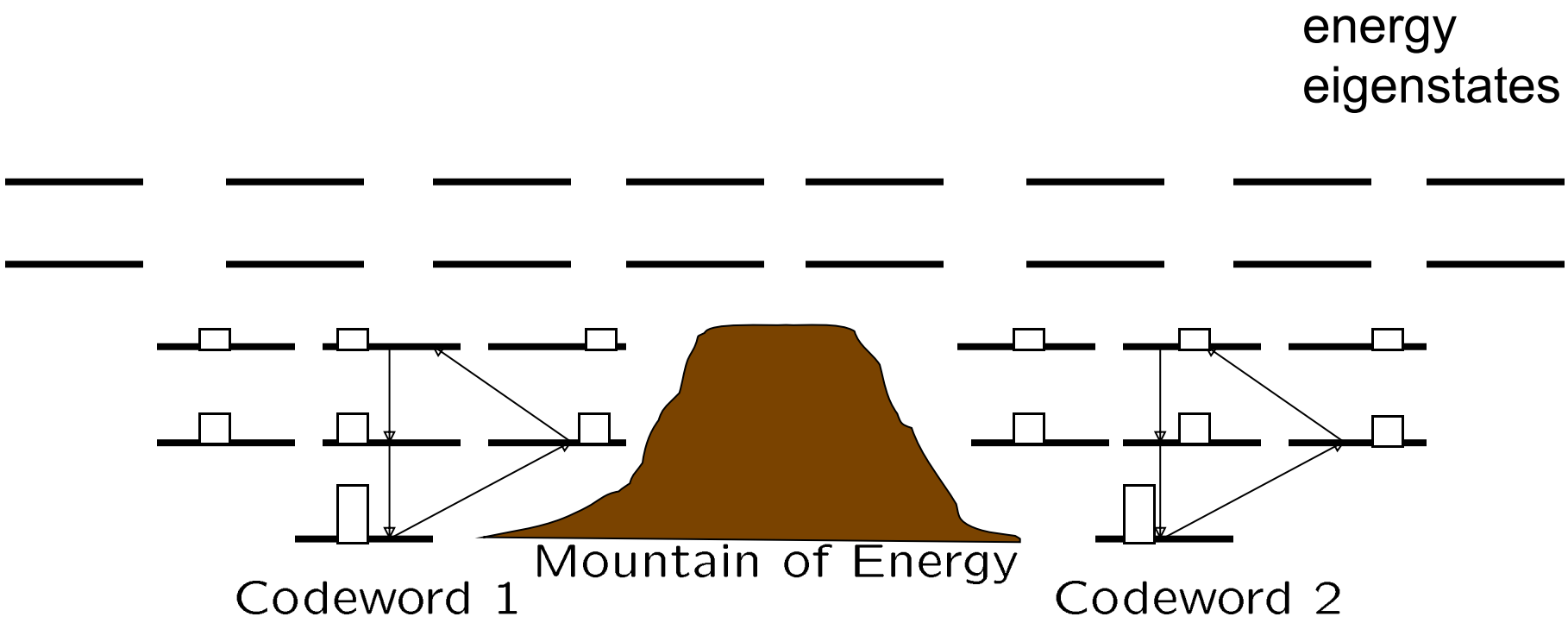
Bacon, Brown, Whaley (2001)

Quantum Error Correcting Is Sophisticated Cooling



Self-Correction

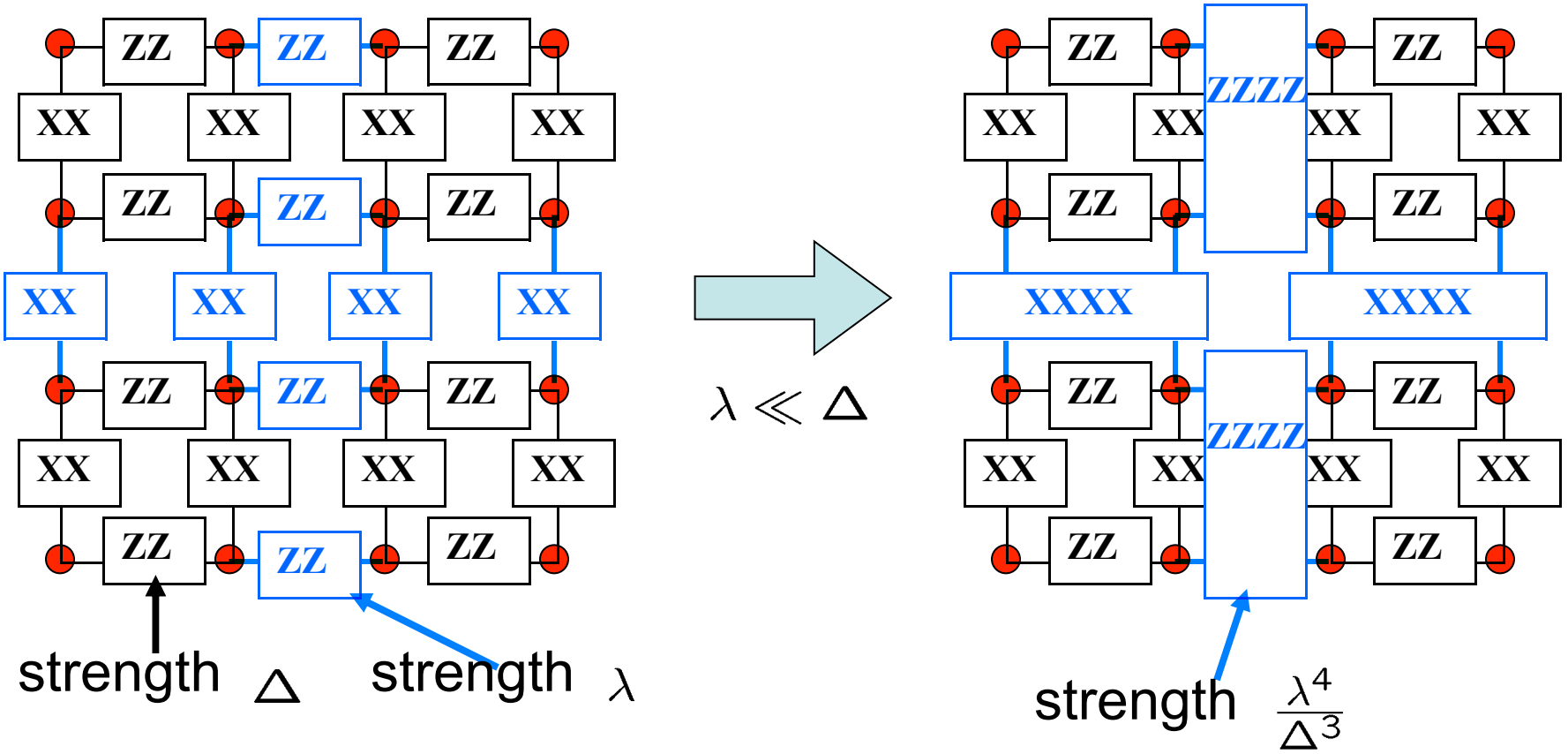
[Dennis et al (2001), Bacon (2006)]

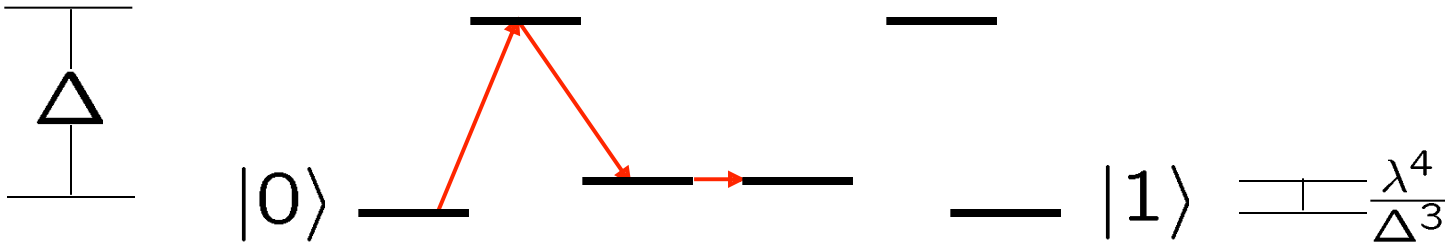


Important for high T operation as well as for fault-tolerance

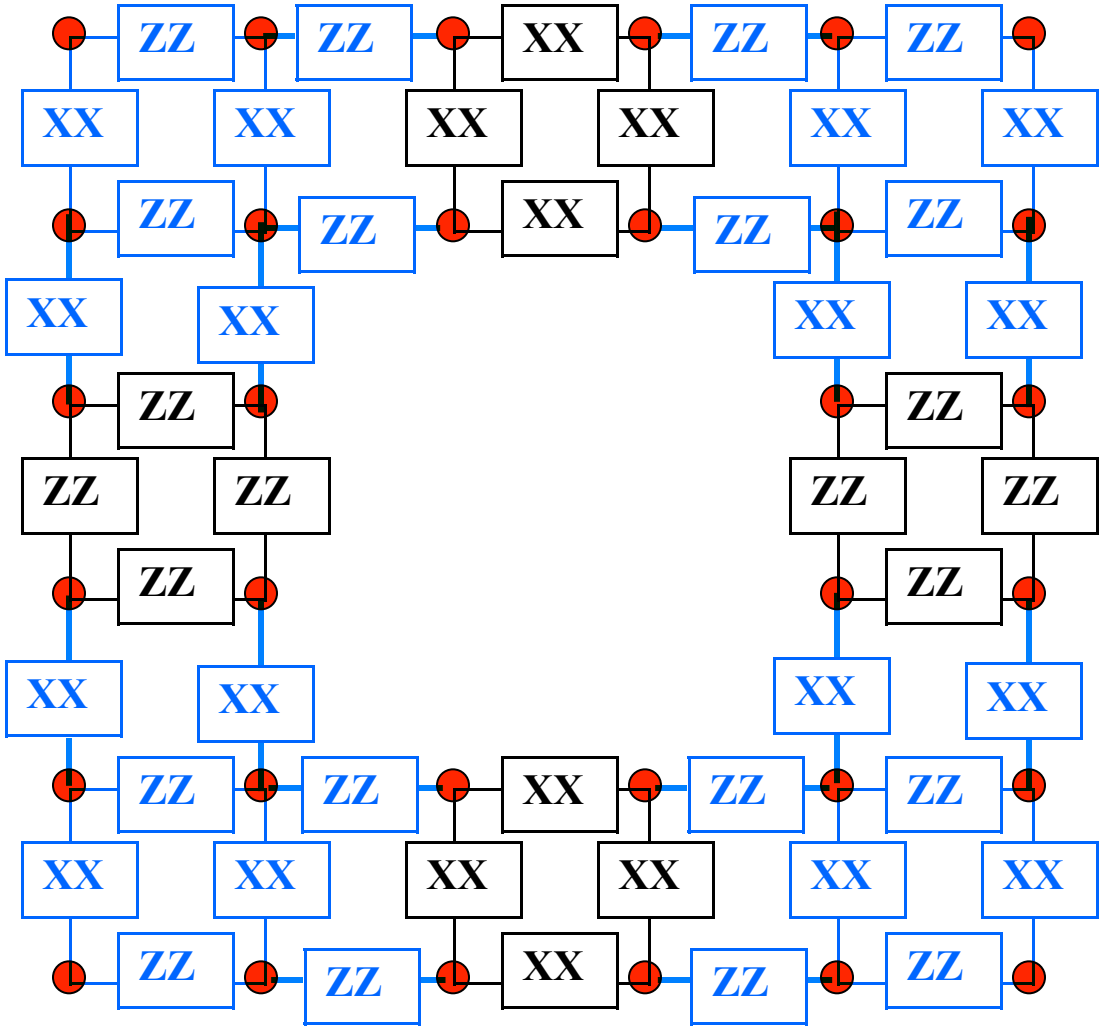
Energy gap is not enough to guarantee fault-tolerance
(though it is sufficient for memory)

Concatenate?



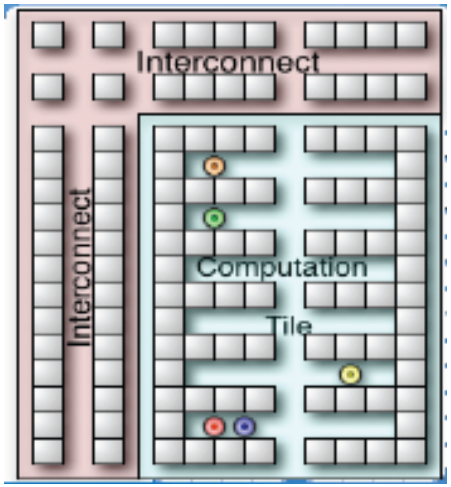


The Proper Concatenation

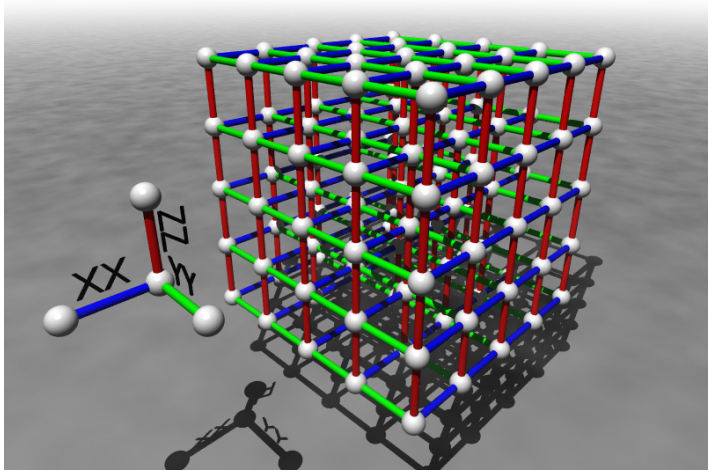


Use gadgets to get strong concatenation: self-correcting!

Who Will Build a Quantum Computer?



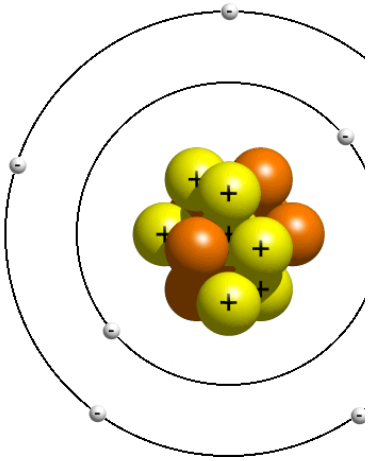
quantum error correcting architectures



physics motivated methods for error correction

Challenge: quantum statistical physics theory of fault-tolerant quantum computing

When Physics and Computer Science Collide







What Worth, Quantum Computation?

Quantum algorithms research is about much more than just coming up with new quantum algorithms.

Bold Reasons

Suppose that tomorrow someone proves that P≠NP, would we be satisfied? BQP≠NP?

The language of the universe is quantum mechanical, shouldn't the language of our theory of computation match?

[Kerenidis, de Wolf 02] Length bounds on 2 query locally decodable codes via a quantum proof

[Aaronson 04] PP closed under intersection via post selected quantum computation

[Kerenidis 05] Classical circuit lower bounds from quantum communication complexity bounds

[Feige] Parallel repetition theorem bounds were done using quantum theory (without the author know about it!)

What Worth, Quantum Computation?

Quantum algorithms research is about much more than just coming up with new quantum algorithms.

Bold Reasons

Understanding what “powers” quantum algorithms helps us come up with new algorithms, but it also helps us understand when a quantum system can be simulated

Entanglement is necessary for quantum speedups

[Jozsa and Linden 02]

New (classical) algorithms for efficient simulation of certain many-body quantum systems

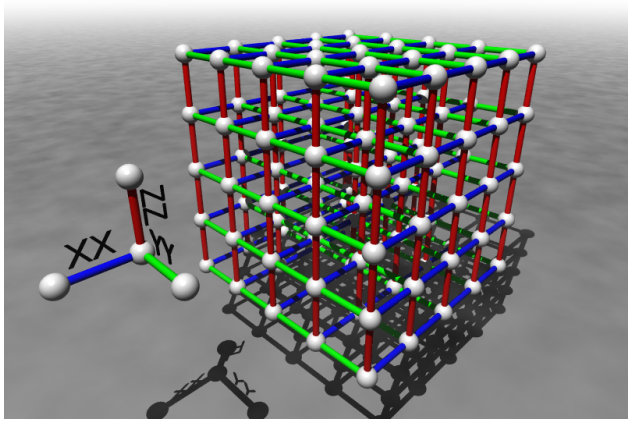
[Vidal 03, Verstraete, Cirac 05]

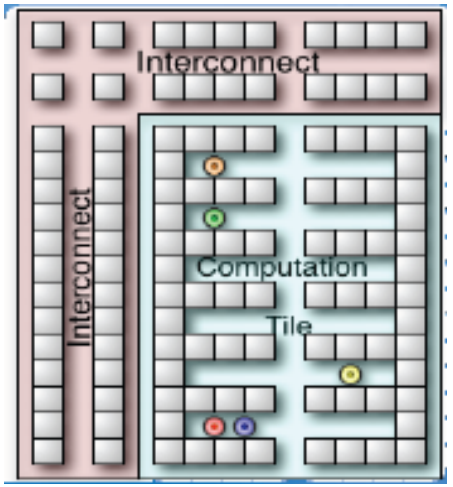
When Physics and Computer Science Collide

Best comment that received the most laughter last week at FOCS: Leslie Valiant in his great talk on 'accidental algorithms': "We have all these complexity classes, and we don't know at all if they are different... It would be a nightmare if they turn out to be all the same, such that NC2=P=P^#P and so on." <comment from the audience about if this would really be so bad> "Yes it would be a nightmare, but the *worst* possible nightmare would be if NC=P=P^#P and if it was a *physicist* who proved it."

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